

1. Mobility Access Index

This KPI measures how well each ward in the city is served by mobility infrastructure. It combines road network density with the availability of ride-hailing services, adjusted for local population. A higher score means the ward is well-connected, while a lower score highlights areas that need infrastructure investment. It is built from the demographic and ride-hailing datasets.

1.2 Significance

- Identifies mobility-poor wards that city-wide averages would otherwise mask.
- Supports equity-focused infrastructure investment planning.
- Provides a baseline to measure future improvement against.

1.3 Components & Data Used

Component	Table	Column
Road infrastructure	Dim_Demographics	Road length (kms), Area (sq km)
Service availability	Fact_RideHailing	Bookings
Population	Dim_Demographics	Female Population, Male Population

1.4 Measures

Road Density per SqKm

Road Density per SqKm =

```
DIVIDE(  
    SUM(Dim_Demographics[Road length (kms)]),  
    SUM(Dim_Demographics[Area (sq km)])  
)
```

RideHailing Availability per 1000Pop

RideHailing Availability per 1000Pop =

```
DIVIDE(  
    SUM(Fact_RideHailing[Bookings]) * 1000,  
    SUM(Dim_Demographics[Female Population]) + SUM(Dim_Demographics[Male  
Population])  
)
```

Norm Road Density

Norm Road Density =

VAR CurrVal = [Road Density per SqKm]

VAR MinVal = CALCULATE([Road Density per SqKm], ALL(Dim_Demographics[Ward Name]))

VAR MaxVal = CALCULATE([Road Density per SqKm], ALLSELECTED(Dim_Demographics[Ward Name]))

RETURN DIVIDE(CurrVal - MinVal, MaxVal - MinVal) * 100

1.5 Final Measure

Mobility Access Index

Mobility Access Index =

[Norm Road Density] * 0.5+ [Norm RideHailing Availability] * 0.5

1.5 Interpretation

Score Range	Meaning
70–100	Well-connected ward — strong roads and service availability
40–69	Moderate access
0–39	Underserved ward — priority for investment

2. City Mobility Health Score (CMHS)

This KPI reflects how well the transport system is functioning at any given time. It combines traffic congestion levels, average vehicle speed, incident frequency, and ride-hailing trip completion rates into a single score. A high score indicates smooth, reliable traffic flow, while a declining score signals emerging congestion or service issues. It is built from the traffic and ride-hailing datasets.

2.2 Significance

- Acts as an early-warning indicator for emerging congestion or service issues.
- Enables comparison across time periods (weekday vs weekend, seasons).
- Gives authorities one figure to report instead of many disconnected charts.

2.3 Components & Data Used

Component	Table	Column
Traffic flow	Fact_Traffic	Average Speed, Congestion Level
Incident frequency	Fact_Traffic	Incident Reports

Component	Table	Column
Service completion	Fact_RideHailing	Completed Trips, Bookings

2.4 Measures

Avg Congestion Level

Avg Congestion Level = AVERAGE(Fact_Traffic[Congestion Level])

Avg Traffic Speed

Avg Traffic Speed = AVERAGE(Fact_Traffic[Average Speed])

Incident Rate

Incident Rate = DIVIDE(SUM(Fact_Traffic[Incident Reports]),
DISTINCTCOUNT(Fact_Traffic[Date]))

Ride Completion Rate

Ride Completion Rate = DIVIDE(SUM(Fact_RideHailing[Completed Trips]),
SUM(Fact_RideHailing[Bookings]))

2.5 Final Measure

City Mobility Health Score

City Mobility Health Score =

VAR CongestionScore = 100 - ([Avg Congestion Level] * 10)

VAR SpeedScore = MIN(100, [Avg Traffic Speed] * 2)

VAR IncidentScore = 100 - MIN(100, [Incident Rate] * 20)

VAR CompletionScore = [Ride Completion Rate] * 100

RETURN CongestionScore*0.3 + SpeedScore*0.2 + IncidentScore*0.25 +
CompletionScore*0.25

3. Reliability Metrics

These metrics track how dependable ride-hailing services are from a commuter's perspective. The Cancellation Rate shows what share of bookings are cancelled, while the Completion Rate shows what share are successfully fulfilled. Together, they reveal whether high demand is actually translating into good service. These are calculated entirely from the ride-hailing dataset.

3.2 Significance

- Reflects commuter experience directly — high demand with poor reliability still means a failing service.
- Separates demand-side issues (driver shortage) from supply-side issues (cancellations).
- A leading indicator of user trust and retention.

3.3 Components & Data Used

Component	Table	Column
Cancelled bookings	Fact_RideHailing	Cancelled Bookings, Bookings
Completed bookings	Fact_RideHailing	Completed Trips, Bookings

3.4 Measures

Ride Cancellation Rate

Ride Cancellation Rate = $\text{DIVIDE}(\text{SUM}(\text{Fact_RideHailing}[\text{Cancelled Bookings}]), \text{SUM}(\text{Fact_RideHailing}[\text{Bookings}]))$

Trip Completion Rate

Trip Completion Rate = $\text{DIVIDE}(\text{SUM}(\text{Fact_RideHailing}[\text{Completed Trips}]), \text{SUM}(\text{Fact_RideHailing}[\text{Bookings}]))$

4. Demand Metrics

These metrics capture how much people are using ride-hailing and bike-sharing services, and when. Total bookings and rides show overall usage volume, bookings per capita allow fair comparison across wards, and peak-hour demand highlights the busiest times of day. This helps with planning fleet size and resource allocation, using the ride-hailing, bike-sharing, and demographic datasets.

4.2 Significance

- Informs fleet sizing and driver/bike allocation by time of day.
- Reveals peak-hour stress points needing targeted intervention.
- Per-capita normalization allows fair comparison between high- and low-population wards.

4.3 Components & Data Used

Component	Table	Column
Ride-hailing volume	Fact_RideHailing	Bookings
Population (per-capita)	Dim_Demographics	Female Population, Male Population
Bike-share volume	Fact_BikeSharing	cnt
Peak-hour pattern	Fact_BikeSharing	hr, cnt

4.4 Measures

Total RideHailing Bookings

Total RideHailing Bookings = SUM(Fact_RideHailing[Bookings])

Bookings per Capita

Bookings per Capita = DIVIDE(SUM(Fact_RideHailing[Bookings]),
SUM(Dim_Demographics[Female Population]) + SUM(Dim_Demographics[Male
Population]))

Total BikeShare Rides

Total BikeShare Rides = SUM(Fact_BikeSharing[cnt])

Peak Hour Demand

Peak Hour Demand = CALCULATE(SUM(Fact_BikeSharing[cnt]), Fact_BikeSharing[hr] IN
{8,9,17,18,19})